

The Double Cover $SU(2) \rightarrow SO(3)$

Road map

The purpose of this chapter is to construct, from first principles, a surjective Lie group homomorphism

$$R : SU(2) \longrightarrow SO(3)$$

whose kernel is $\{I_2, -I_2\}$. Thus every three-dimensional rotation has exactly two preimages in $SU(2)$, and

$$SO(3) \cong SU(2)/\{\pm I_2\}.$$

We shall:

1. identify $\mathfrak{su}(2)$ with a three-dimensional real inner product space using the Pauli matrices;
2. let $SU(2)$ act on this space by conjugation and write that action as a 3×3 real matrix $R(u)$;
3. verify directly that R is continuous, multiplicative, and orthogonal with determinant 1;
4. compute the differential $\mathfrak{su}(2) \rightarrow \mathfrak{so}(3)$, obtaining the familiar cross-product generators of rotations;
5. prove surjectivity and compute the kernel;
6. explain why $SU(2)$ is the three-sphere, while $SO(3)$ is a closed three-ball with antipodal boundary points identified.

18.1 Pauli matrices and the real space $\mathfrak{su}(2)$

Recall the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They are Hermitian and traceless. Their basic multiplication law is

$$\sigma_\alpha \sigma_\beta = \delta_{\alpha\beta} I_2 + i \sum_{\gamma=1}^3 \varepsilon_{\alpha\beta\gamma} \sigma_\gamma, \quad (18.1)$$

where $\varepsilon_{\alpha\beta\gamma}$ is the Levi–Civita symbol: it is $+1$ for even permutations of 123 , -1 for odd permutations, and 0 when two indices coincide. Consequently,

$$[\sigma_\alpha, \sigma_\beta] = 2i \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} \sigma_\gamma, \quad (18.2)$$

$$\text{tr}(\sigma_\alpha \sigma_\beta) = 2\delta_{\alpha\beta}. \quad (18.3)$$

The Lie algebra of $SU(2)$ is

$$\mathfrak{su}(2) = \left\{ X \in \mathfrak{gl}(2, \mathbb{C}) : X^\dagger = -X, \quad \text{tr } X = 0 \right\}.$$

Every $X \in \mathfrak{su}(2)$ has a unique expression

$$X = i \sum_{\alpha=1}^3 y_\alpha \sigma_\alpha = i \mathbf{y} \cdot \boldsymbol{\sigma}, \quad \mathbf{y} = (y_1, y_2, y_3) \in \mathbb{R}^3. \quad (18.4)$$

Indeed, a general anti-Hermitian traceless 2×2 matrix is

$$\begin{pmatrix} iy_3 & y_2 + iy_1 \\ -y_2 + iy_1 & -iy_3 \end{pmatrix} = iy_1 \sigma_1 + iy_2 \sigma_2 + iy_3 \sigma_3.$$

Thus

$$\dim_{\mathbb{R}} \mathfrak{su}(2) = 3, \quad \{i\sigma_1, i\sigma_2, i\sigma_3\}$$

is a real basis.

18.1.1 The normalized Frobenius inner product

On the real vector space $\mathfrak{su}(2)$, define

$$\langle X, Y \rangle = \frac{1}{2} \text{tr}(X^\dagger Y). \quad (18.5)$$

This is the suitably normalized Frobenius inner product. By (18.3),

$$\langle i\sigma_\alpha, i\sigma_\beta \rangle = \frac{1}{2} \text{tr}((i\sigma_\alpha)^\dagger i\sigma_\beta) = \frac{1}{2} \text{tr}(\sigma_\alpha \sigma_\beta) = \delta_{\alpha\beta}.$$

Hence the basis $\{i\sigma_\alpha\}$ is orthonormal. Therefore, if

$$Z = \sum_{\alpha=1}^3 z_\alpha i\sigma_\alpha,$$

then its coordinates are recovered by

$$z_\alpha = \langle i\sigma_\alpha, Z \rangle.$$

18.2 Constructing the map by the adjoint action

For $u \in SU(2)$, conjugation preserves anti-Hermiticity and trace:

$$(uXu^\dagger)^\dagger = uX^\dagger u^\dagger = -uXu^\dagger, \quad \text{tr}(uXu^\dagger) = \text{tr } X.$$

Thus

$$\text{Ad}_u : \mathfrak{su}(2) \longrightarrow \mathfrak{su}(2), \quad X \longmapsto uXu^{-1} = uXu^\dagger,$$

is a real-linear automorphism.

Definition 18.1: The rotation matrix associated with u

For $u \in SU(2)$, let $R(u)$ be the matrix of Ad_u in the ordered orthonormal basis $\{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3\}$. Thus

$$u(\mathbf{i}\sigma_\beta)u^\dagger = \sum_{\alpha=1}^3 R(u)_{\alpha\beta} \mathbf{i}\sigma_\alpha. \quad (18.6)$$

Taking the inner product with $\mathbf{i}\sigma_\alpha$ gives the explicit coordinate formula appearing in the handwritten derivation:

$$\begin{aligned} R(u)_{\alpha\beta} &= \langle \mathbf{i}\sigma_\alpha, u(\mathbf{i}\sigma_\beta)u^\dagger \rangle \\ &= \frac{1}{2} \text{tr}((\mathbf{i}\sigma_\alpha)^\dagger u(\mathbf{i}\sigma_\beta)u^\dagger) \\ &= \boxed{\frac{1}{2} \text{tr}(\sigma_\alpha u \sigma_\beta u^\dagger)}. \end{aligned} \quad (18.7)$$

Although complex matrices occur on the right, the result is real because it is the coordinate of an element of the real vector space $\mathfrak{su}(2)$ in a real basis.

18.2.1 The homomorphism property, index by index

Let $u_1, u_2 \in SU(2)$. Starting from the defining expansion,

$$\begin{aligned} (u_1 u_2)(\mathbf{i}\sigma_\beta)(u_1 u_2)^\dagger &= u_1 (u_2(\mathbf{i}\sigma_\beta)u_2^\dagger) u_1^\dagger \\ &= u_1 \left(\sum_{\mu} R(u_2)_{\mu\beta} \mathbf{i}\sigma_\mu \right) u_1^\dagger \\ &= \sum_{\mu} R(u_2)_{\mu\beta} \sum_{\alpha} R(u_1)_{\alpha\mu} \mathbf{i}\sigma_\alpha \\ &= \sum_{\alpha} \left(\sum_{\mu} R(u_1)_{\alpha\mu} R(u_2)_{\mu\beta} \right) \mathbf{i}\sigma_\alpha. \end{aligned}$$

Uniqueness of coordinates yields

$$R(u_1 u_2)_{\alpha\beta} = \sum_{\mu} R(u_1)_{\alpha\mu} R(u_2)_{\mu\beta}.$$

In matrix form,

$$\boxed{R(u_1 u_2) = R(u_1) R(u_2)}. \quad (18.8)$$

Also,

$$R(I_2) = I_3, \quad R(u^{-1}) = R(u)^{-1}.$$

18.2.2 Why $R(u)$ is orthogonal

Conjugation by a unitary matrix preserves the Frobenius inner product:

$$\begin{aligned} \langle uXu^\dagger, uYu^\dagger \rangle &= \frac{1}{2} \operatorname{tr}((uXu^\dagger)^\dagger uYu^\dagger) \\ &= \frac{1}{2} \operatorname{tr}(uX^\dagger u^\dagger uYu^\dagger) \\ &= \frac{1}{2} \operatorname{tr}(uX^\dagger Yu^\dagger) \\ &= \frac{1}{2} \operatorname{tr}(X^\dagger Y) = \langle X, Y \rangle. \end{aligned}$$

Applying this to the orthonormal basis vectors gives

$$\begin{aligned} \delta_{\alpha\beta} &= \langle u(i\sigma_\alpha)u^\dagger, u(i\sigma_\beta)u^\dagger \rangle \\ &= \sum_{\gamma, \mu} R(u)_{\gamma\alpha} R(u)_{\mu\beta} \langle i\sigma_\gamma, i\sigma_\mu \rangle \\ &= \sum_{\gamma} R(u)_{\gamma\alpha} R(u)_{\gamma\beta}. \end{aligned}$$

Therefore

$$R(u)^\top R(u) = I_3, \quad R(u) \in O(3). \quad (18.9)$$

In particular,

$$R(u^\dagger) = R(u)^{-1} = R(u)^\top,$$

which also follows directly from the trace formula:

$$R(u^\dagger)_{\alpha\beta} = \frac{1}{2} \operatorname{tr}(\sigma_\alpha u^\dagger \sigma_\beta u) = R(u)_{\beta\alpha}.$$

18.2.3 Continuity

Every entry

$$R(u)_{\alpha\beta} = \frac{1}{2} \operatorname{tr}(\sigma_\alpha u \sigma_\beta u^\dagger)$$

is a polynomial expression in the real and imaginary parts of the entries of u . Hence every matrix entry is continuous, so

$$R : SU(2) \longrightarrow O(3)$$

is continuous. Together with (18.8), this proves that R is a Lie group homomorphism once we establish that its image lies in the component $SO(3)$. We shall do so explicitly after computing its differential.

18.3 The induced Lie-algebra homomorphism

Let

$$r : \mathfrak{su}(2) \longrightarrow \mathfrak{so}(3)$$

be the differential of R at the identity. By definition,

$$r(X) = \left. \frac{d}{dt} R(e^{tX}) \right|_{t=0}. \quad (18.10)$$

Since R is the matrix representation of the adjoint action, the compatibility of homomorphisms with exponentials gives

$$R(e^X) = \text{the matrix of } \text{Ad}_{e^X} = \text{the matrix of } e^{\text{ad}_X}.$$

Therefore $r(X)$ is precisely the matrix of ad_X in the basis $\{\mathbf{i}\sigma_\alpha\}$.

For completeness, differentiate the entries directly:

$$\begin{aligned} r(X)_{\alpha\beta} &= \left. \frac{d}{dt} \frac{1}{2} \text{tr}(\sigma_\alpha e^{tX} \sigma_\beta e^{-tX}) \right|_{t=0} \\ &= \frac{1}{2} \text{tr}(\sigma_\alpha (X\sigma_\beta - \sigma_\beta X)) \\ &= \langle \mathbf{i}\sigma_\alpha, [X, \mathbf{i}\sigma_\beta] \rangle. \end{aligned} \quad (18.11)$$

18.3.1 Cross-product matrices and the sign convention

For $\mathbf{y} \in \mathbb{R}^3$, define $T(\mathbf{y}) \in \mathfrak{so}(3)$ by

$$T(\mathbf{y})\mathbf{v} = \mathbf{y} \times \mathbf{v}.$$

Writing

$$T(\mathbf{y}) = y_1 T_1 + y_2 T_2 + y_3 T_3,$$

the three generators are

$$T_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad T_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (18.12)$$

Thus

$$T(\mathbf{y}) = \begin{pmatrix} 0 & -y_3 & y_2 \\ y_3 & 0 & -y_1 \\ -y_2 & y_1 & 0 \end{pmatrix}.$$

Let $X = \mathbf{i} \mathbf{y} \cdot \boldsymbol{\sigma}$. From (18.2),

$$\begin{aligned} [X, \mathbf{i}\sigma_\beta] &= \sum_{\gamma} y_{\gamma} [\mathbf{i}\sigma_{\gamma}, \mathbf{i}\sigma_{\beta}] \\ &= -2 \sum_{\gamma, \alpha} y_{\gamma} \varepsilon_{\gamma\beta\alpha} \mathbf{i}\sigma_{\alpha}. \end{aligned}$$

Since

$$T(\mathbf{y})_{\alpha\beta} = \sum_{\gamma} \varepsilon_{\gamma\beta\alpha} y_{\gamma},$$

equation (18.11) gives

$$r(\mathbf{i} \mathbf{y} \cdot \boldsymbol{\sigma}) = -2T(\mathbf{y}). \quad (18.13)$$

The minus sign is not mysterious: it comes from choosing $\mathbf{i}\sigma_{\alpha}$, rather than $-\mathbf{i}\sigma_{\alpha}$, as the basis of $\mathfrak{su}(2)$, together with the convention $T(\mathbf{y})\mathbf{v} = \mathbf{y} \times \mathbf{v}$.

Proposition 18.1: The Lie algebras are isomorphic

The differential r is an isomorphism

$$\mathfrak{su}(2) \cong \mathfrak{so}(3).$$

With

$$e_{\alpha} = \mathbf{i}\sigma_{\alpha}, \quad f_{\alpha} = -2T_{\alpha},$$

we have $r(e_{\alpha}) = f_{\alpha}$, and both bases have the same structure constants:

$$[e_{\alpha}, e_{\beta}] = -2 \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} e_{\gamma}, \quad [f_{\alpha}, f_{\beta}] = -2 \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} f_{\gamma}.$$

证明. The first bracket formula follows from (18.2). The cross-product generators satisfy

$$[T_{\alpha}, T_{\beta}] = \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} T_{\gamma}.$$

Hence

$$\begin{aligned} [-2T_{\alpha}, -2T_{\beta}] &= 4[T_{\alpha}, T_{\beta}] \\ &= 4 \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} T_{\gamma} \\ &= -2 \sum_{\gamma} \varepsilon_{\alpha\beta\gamma} (-2T_{\gamma}). \end{aligned}$$

Thus $e_{\alpha} \mapsto f_{\alpha}$ preserves brackets. It maps a basis of the three-dimensional space $\mathfrak{su}(2)$ to a basis of the three-dimensional space $\mathfrak{so}(3)$, so it is an isomorphism. ■

18.4 The explicit rotation generated by u

18.4.1 Every element of $SU(2)$ is an exponential

A matrix $u \in SU(2)$ has the form

$$u = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}, \quad |a|^2 + |b|^2 = 1. \quad (18.14)$$

To see this, write the first row as (a, b) . A unit vector orthogonal to it is $(-b^*, a^*)$; the determinant-one condition fixes the otherwise possible unit complex phase.

Write

$$a = a_0 + ia_3, \quad b = a_2 + ia_1, \quad a_j \in \mathbb{R}.$$

Then

$$u = a_0 I_2 + i \sum_{\alpha=1}^3 a_\alpha \sigma_\alpha, \quad a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1. \quad (18.15)$$

Choose $s \in [0, \pi]$ such that $a_0 = \cos s$. When $\sin s \neq 0$, put

$$\mathbf{n} = \frac{(a_1, a_2, a_3)}{\sin s}, \quad \|\mathbf{n}\| = 1.$$

At $\sin s = 0$, any unit $\mathbf{n}$ may be used. Because

$$(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I_2,$$

its powers separate into even and odd terms, and therefore

$$\begin{aligned} \exp(is \mathbf{n} \cdot \boldsymbol{\sigma}) &= \sum_{k=0}^{\infty} \frac{(is \mathbf{n} \cdot \boldsymbol{\sigma})^k}{k!} \\ &= \cos s I_2 + i \sin s \mathbf{n} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (18.16)$$

Thus every $u \in SU(2)$ can be written

$$u = e^{is \mathbf{n} \cdot \boldsymbol{\sigma}}.$$

18.4.2 Exponentiating the differential

Let

$$u = e^{iy \cdot \boldsymbol{\sigma}}, \quad \mathbf{y} = s\mathbf{n}, \quad \|\mathbf{n}\| = 1.$$

The exponential compatibility of a Lie group homomorphism, together with (18.13), gives

$$\boxed{R(u) = R(e^{iy \cdot \boldsymbol{\sigma}}) = e^{-2T(\mathbf{y})} = e^{-2sT(\mathbf{n})}.} \quad (18.17)$$

For a unit vector $\mathbf{n}$,

$$T(\mathbf{n})^2 = \mathbf{n}\mathbf{n}^T - I_3, \quad T(\mathbf{n})^3 = -T(\mathbf{n}).$$

The exponential series then gives Rodrigues' formula

$$\begin{aligned} e^{\theta T(\mathbf{n})} &= I_3 + \sin \theta T(\mathbf{n}) + (1 - \cos \theta) T(\mathbf{n})^2 \\ &= \cos \theta I_3 + (1 - \cos \theta) \mathbf{n}\mathbf{n}^T + \sin \theta T(\mathbf{n}). \end{aligned} \quad (18.18)$$

It fixes $\mathbf{n}$ and rotates the plane $\mathbf{n}^\perp$ through angle θ , with the right-hand-rule orientation. Therefore

$$R(e^{is \mathbf{n} \cdot \boldsymbol{\sigma}})$$

is the rotation about the axis $\mathbf{n}$ through angle $-2s$. Equivalently, the conventional rotation through angle θ about $\mathbf{n}$ is obtained from either of the two matrices

$$u_{\pm} = \pm \exp\left(-\frac{i\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right). \quad (18.19)$$

Conjugation identity in vector form

Equation (18.6) is equivalent to

$$u \mathbf{i}(\mathbf{v} \cdot \boldsymbol{\sigma}) u^\dagger = \mathbf{i}((R(u)\mathbf{v}) \cdot \boldsymbol{\sigma}) \quad (\mathbf{v} \in \mathbb{R}^3).$$

Thus conjugating a traceless anti-Hermitian 2×2 matrix by u performs an ordinary three-dimensional rotation on its coefficient vector.

18.4.3 The determinant is 1

We already know that $R(u) \in O(3)$. Formula (18.17) gives the orientation directly:

$$\begin{aligned} \det R(u) &= \det(e^{-2T(\mathbf{y})}) \\ &= \exp(\operatorname{tr}(-2T(\mathbf{y}))) \\ &= e^0 = 1, \end{aligned}$$

because every skew-symmetric matrix has zero diagonal and hence zero trace. Consequently,

$$R(u) \in SO(3) \quad (u \in SU(2)). \quad (18.20)$$

This completes the verification that

$$R : SU(2) \longrightarrow SO(3)$$

is a continuous group homomorphism.

18.5 Surjectivity and the two-element kernel**18.5.1 Every rotation has a preimage****Proposition 18.2: Surjectivity**

The homomorphism

$$R : SU(2) \longrightarrow SO(3)$$

is onto.

证明. Every $Q \in SO(3)$ has a real eigenvector $\mathbf{n}$ with eigenvalue 1. Indeed, a real cubic characteristic polynomial has a real root, and the eigenvalues of an orthogonal matrix have absolute value 1. Nonreal eigenvalues occur in conjugate pairs whose product is 1; since $\det Q = 1$, the remaining real eigenvalue must be 1.

The orthogonal plane $\mathbf{n}^\perp$ is Q -invariant, and the restriction of Q to that oriented plane is a two-dimensional rotation through some angle θ . Thus, by Rodrigues' formula,

$$Q = e^{\theta T(\mathbf{n})}.$$

Set

$$u = \exp\left(-\frac{\mathbf{i}\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right) \in SU(2).$$

Equation (18.17) gives $R(u) = Q$. ■

18.5.2 Computing the kernel from the exponential formula

Let

$$u = e^{is\mathbf{n} \cdot \boldsymbol{\sigma}}, \quad \|\mathbf{n}\| = 1.$$

The skew-symmetric matrix $T(\mathbf{n})$ has eigenvalues

$$0, \quad i, \quad -i.$$

The zero eigenvalue corresponds to the rotation axis $\mathbf{n}$; on $\mathbf{n}^\perp$, the matrix is a quarter-turn. Hence the eigenvalues of

$$R(u) = e^{-2sT(\mathbf{n})}$$

are

$$1, \quad e^{-2is}, \quad e^{2is}.$$

Therefore

$$R(u) = I_3 \iff e^{2is} = 1 \iff s = m\pi \quad (m \in \mathbb{Z}).$$

Using (18.16),

$$u = \cos(m\pi)I_2 + i \sin(m\pi)\mathbf{n} \cdot \boldsymbol{\sigma} = (-1)^m I_2.$$

Thus

$$\ker R = \{I_2, -I_2\}. \tag{18.21}$$

A second, basis-free kernel check

If $R(u) = I_3$, then

$$u\sigma_\alpha u^\dagger = \sigma_\alpha \quad (\alpha = 1, 2, 3),$$

so u commutes with all Pauli matrices. Commutation with σ_3 forces u to be diagonal, and commutation with σ_1 forces its two diagonal entries to be equal. Hence $u = \lambda I_2$. Unitarity gives $|\lambda| = 1$, while $\det u = \lambda^2 = 1$; therefore $\lambda = \pm 1$.

Theorem 18.1: $SU(2)$ double-covers $SO(3)$

The adjoint construction gives a surjective Lie group homomorphism

$$R : SU(2) \longrightarrow SO(3)$$

with kernel $\{\pm I_2\}$. It induces a Lie group isomorphism

$$SU(2)/\{\pm I_2\} \cong SO(3). \tag{18.22}$$

Moreover, R is a two-sheeted covering map.

证明. Surjectivity and the kernel have already been proved. The first isomorphism theorem gives the quotient isomorphism.

To see the covering property concretely, note that the differential

$$r : \mathfrak{su}(2) \longrightarrow \mathfrak{so}(3)$$

is an isomorphism. The inverse function theorem therefore provides a neighbourhood U of I_2 on which R is a diffeomorphism onto a neighbourhood V of I_3 . Shrink U , if necessary, so that $U \cap (-U) = \emptyset$.

If $g \in R^{-1}(V)$, there is a unique $u \in U$ with $R(u) = R(g)$. Then

$$R(gu^{-1}) = I_3,$$

so $gu^{-1} = \pm I_2$, or $g = \pm u$. Consequently,

$$R^{-1}(V) = U \sqcup (-U),$$

and R maps each of the two pieces diffeomorphically onto V . Left translation gives the same description over a neighbourhood of every point of $SO(3)$. Hence R is a two-sheeted covering. ■

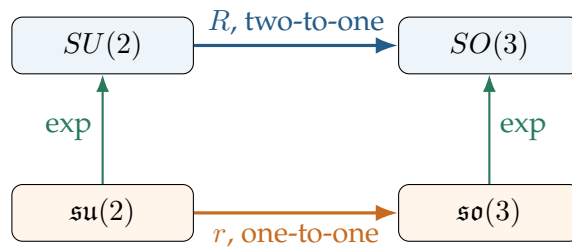


图 18.1: The group map is two-to-one, although its differential is an isomorphism. The square commutes: $R(e^X) = e^{r(X)}$.

18.6 Axis-angle coordinates and the topology of $SO(3)$

Every element of $SO(3)$ can be described by a unit axis $\mathbf{n} \in S^2$ and an angle $\theta \in [0, \pi]$. Associate to the rotation the vector

$$\mathbf{x} = \theta \mathbf{n} \in \mathbb{R}^3.$$

All vectors with $\|\mathbf{x}\| < \pi$ describe distinct rotations. On the boundary $\|\mathbf{x}\| = \pi$, however,

$$(\mathbf{n}, \pi) \quad \text{and} \quad (-\mathbf{n}, \pi)$$

describe the same rotation. A half-turn about $-\mathbf{n}$ is the same linear transformation as a half-turn about $\mathbf{n}$.

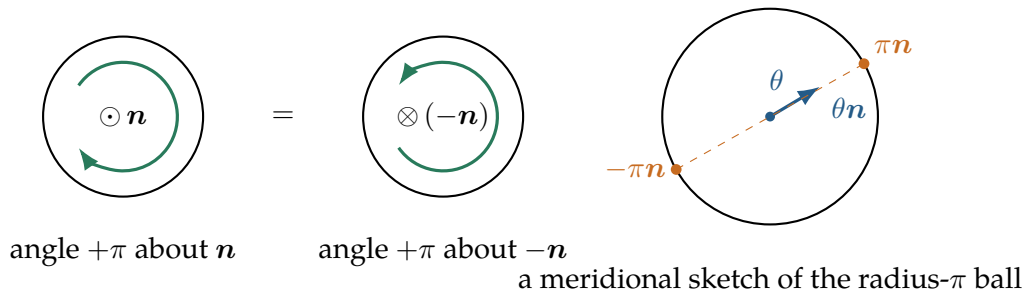


图 18.2: Axis-angle coordinates. Antipodal points on the boundary sphere represent the same rotation by π .

It follows that, topologically,

$$SO(3) \cong B_\pi^3 / (x \sim -x \text{ for } \|x\| = \pi) \cong \mathbb{RP}^3. \quad (18.23)$$

Here B_π^3 is the *three*-dimensional closed ball of radius π . Its two-dimensional drawings are only cross-sectional sketches; the rotation group itself is a three-dimensional manifold, which may be embedded in a higher-dimensional Euclidean space but is not an ordinary solid torus.

18.6.1 Closed paths through the identity

In the ball model, a small closed curve lying in the interior can be continuously shrunk to the origin, which represents the identity rotation. A path that begins at a boundary point πn and ends at its antipode $-\pi n$ becomes a closed path only *after* the boundary identification. It cannot be shrunk to a point in $SO(3)$. Traversing that path twice, however, produces a contractible loop.

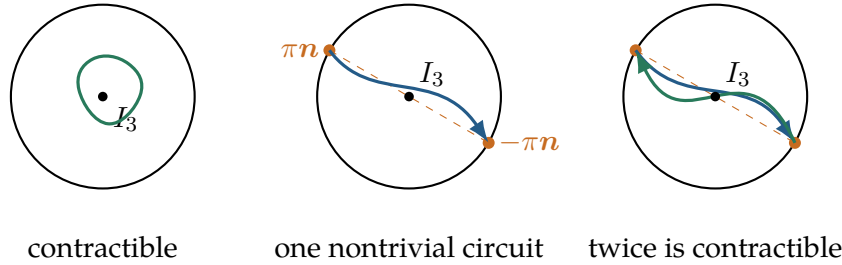


图 18.3: Schematic loops in the antipodally identified ball model of $SO(3)$.

The covering map makes this distinction precise. Consider the loop

$$\gamma(t) = \text{rotation through angle } 2\pi t \text{ about } n, \quad 0 \leq t \leq 1.$$

Its lift to $SU(2)$ beginning at I_2 is

$$\tilde{\gamma}(t) = \exp(-i\pi t n \cdot \sigma).$$

The endpoints are

$$\tilde{\gamma}(0) = I_2, \quad \tilde{\gamma}(1) = -I_2.$$

Thus the lifted path is not closed. A contractible loop has a closed lift when lifted from a chosen point, so γ is not contractible. If the rotation is carried through 4π , the lift ends at

$$\exp(-2\pi i n \cdot \sigma) = I_2;$$

that doubled loop is contractible.

18.7 $SU(2)$ is the three-sphere

The parametrization (18.15) gives a continuous bijection

$$\begin{aligned} S^3 &= \{(a_0, a_1, a_2, a_3) \in \mathbb{R}^4 : a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1\} \\ &\longrightarrow SU(2), \\ (a_0, a_1, a_2, a_3) &\longmapsto a_0 I_2 + i \sum_{\alpha=1}^3 a_\alpha \sigma_\alpha. \end{aligned}$$

Its inverse reads off the four real coefficients and is continuous. Therefore it is a homeomorphism:

$$\boxed{SU(2) \cong S^3.} \quad (18.24)$$

The sphere S^3 is simply connected. One standard proof covers it by the complements of the north and south poles. Each complement is stereographically homeomorphic to $\mathbb{R}^3$, hence is simply connected, and their intersection is path-connected. The Seifert–van Kampen theorem then gives

$$\pi_1(S^3) = \{1\}.$$

Consequently $SU(2)$ is simply connected.

Because $SU(2)$ is simply connected and

$$R : SU(2) \longrightarrow SO(3)$$

is a two-sheeted covering, $SU(2)$ is the universal covering group of $SO(3)$. Its deck transformation group is

$$\{u \mapsto u, u \mapsto -u\} \cong \mathbb{Z}_2.$$

Equivalently,

$$\boxed{\pi_1(SO(3)) \cong \mathbb{Z}_2, \quad \pi_1(SU(2)) = \{1\}.} \quad (18.25)$$

Why identical Lie algebras do not imply identical groups

The differential r proves $\mathfrak{su}(2) \cong \mathfrak{so}(3)$, so the two groups have the same local infinitesimal structure. Globally they differ:

$$SU(2) \cong S^3 \quad \text{is simply connected,} \quad SO(3) \cong \mathbb{RP}^3 \quad \text{has fundamental group } \mathbb{Z}_2.$$

Lie algebras determine connected Lie groups only up to covering and discrete central quotients.

What to retain from this chapter

1. The real Lie algebra

$$\mathfrak{su}(2) = \text{span}_{\mathbb{R}}\{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3\}$$

is three-dimensional, and the displayed basis is orthonormal for $\langle X, Y \rangle = \frac{1}{2} \text{tr}(X^\dagger Y)$.

2. Conjugation defines

$$u(\mathbf{i}\sigma_\beta)u^\dagger = \sum_{\alpha} R(u)_{\alpha\beta} \mathbf{i}\sigma_\alpha, \quad R(u)_{\alpha\beta} = \frac{1}{2} \text{tr}(\sigma_\alpha u \sigma_\beta u^\dagger).$$

3. The map $R : SU(2) \rightarrow SO(3)$ is a continuous group homomorphism. Its differential is

$$r(\mathbf{i} \mathbf{y} \cdot \boldsymbol{\sigma}) = -2T(\mathbf{y}), \quad T(\mathbf{y})\mathbf{v} = \mathbf{y} \times \mathbf{v}.$$

Hence $\mathfrak{su}(2) \cong \mathfrak{so}(3)$.

4. If $u = e^{\mathbf{i} s \mathbf{n} \cdot \boldsymbol{\sigma}}$, then

$$R(u) = e^{-2sT(\mathbf{n})}.$$

Thus the angle in $SO(3)$ is twice the corresponding $SU(2)$ half-angle, with the sign fixed by our basis convention.

5. The map is surjective and

$$\ker R = \{\pm I_2\}, \quad SU(2)/\{\pm I_2\} \cong SO(3).$$

It is a two-sheeted covering map.

6. Topologically,

$$SU(2) \cong S^3, \quad SO(3) \cong B_\pi^3 / (\text{antipodal boundary points}) \cong \mathbb{RP}^3.$$

7. Therefore

$$\pi_1(SU(2)) = \{1\}, \quad \pi_1(SO(3)) \cong \mathbb{Z}_2.$$

A 2π rotation represents the nontrivial loop class, whereas a 4π rotation is contractible.